



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B. Sc. Physics Honours				
Course Title	MACHINE LEARNING USING PYTHON				
Type of Course	Major Elective (SPECIALIZATION IV: DATA SCIENCE AND ARTIFICIAL INTELLIGENCE)				
Semester	VI				
Academic Level	300 -399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Awareness of algorithmic approaches 2. Data Analysis Using Python				
Course Summary	This course deals with various algorithms to enable computers to learn data without being explicitly programmed. An insight into various types of machine learning algorithms, strategies for model generation and evaluation are given in this course.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the concepts and importance of Machine Learning, its types	U	C	Instructor-created exams / Quiz
CO2	Understand & apply Scikit-learn module for Machine Learning	U,Ap	P	Instructor-created exams, Observation of Practical Skills

CO3	Understand the supervised learning algorithms and its application	U,Ap	P	Instructor-created exams/ Quiz/Assignment/ Seminar
CO4	Understand the unsupervised learning algorithms and its application	U,Ap	P	Instructor-created exams/ Quiz/Assignment/ Seminar
CO5	Understand the semi supervised learning algorithms and its application	Ap	P	Practical assignments and practical tests
CO6	Develop critical thinking skills to analyze and solve complex problems using machine learning approaches	Ap	P	Practical assignments and practical tests
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create(C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Mark
I	Introduction to Machine Learning		10	15
	1	Introduction to Machine Learning - Learning Types of Machine Learning	2	
	2	Computational Tools for ML	2	
	3	Introduction to Scikit-learn, Getting Datasets, Generating Your Own Dataset	2	
	4	Getting Started with Scikit-learn- Fitting the Model, Making Predictions,	2	
	5	Data Cleansing- Cleaning Rows with NaNs, Removing Duplicate Rows	2	
	References Chapter 9 of Book 1, Chapter 5 of Book 3			
II	Supervised Learning Algorithms		15	25
	6	Supervised Learning – Introduction and Types	1	

	7	Regression - Simple Linear Regression , Multiple Linear Regression	3	
	8	Classification	2	
	9	Naive Bayes classifier algorithm	3	
	10	Decision Tree	2	
	11	K nearest neighbor (KNN)	2	
	12	Logistic Regression	2	
	References Chapter 10,11 of Book 1, Chapter 2 of Book 2			
III	Unsupervised Learning Algorithms		15	18
	13	Unsupervised Learning	2	
	14	Clustering	2	
	15	K-means Clustering	3	
	16	Hierarchical clustering - Two approaches	3	
	17	Association rule learning	2	
	18	Apriori Algorithm	3	
	References Chapter 10,11 of Book 1, Chapter 3 of Book 2			
IV	Reinforcement Learning		8	12
	19	Semi-supervised learning	2	
	20	Markov Decision Process (MDP)	2	
	21	Markov Chain and Markov Process	2	
	22	Applications of Markov Decision Process	2	

	References Chapter 5 of Book 2		
V	Open Ended Module:	12	
Books and References : 1. Data Science and Machine Learning using Python by Reema Thereja Mc Graw Hill (Book 1) 2. Machine Learning by Dr Ruchi Doshi, Dr Kamal Kant Hiran, Ritesh Kumar Jain Dr Kamlesh Lakhwani BPB Publications (Book 2) 3. Python Machine Learning by Wei-Meng Lee , John Wiley & Sons (Book 3) 4. Machine Learning in Data Science using Python, Dr. R. Nageswara Rao, Dream tech press, 2022(Book 4)			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	0	2	0	3	1	3	1	1	2	3	1	1
CO 2	2	2	3	1	3	3	2	2	2	3	2	2	2
CO 3	2	2	3	1	3	3	2	2	2	3	3	2	2
CO 4	2	2	3	1	3	3	2	2	2	3	3	2	2
CO 5	2	2	3	1	3	3	2	2	2	3	3	2	2
CO 6	1	2	3	2	2	3	3	3	3	3	3	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	